Lesson 4
Interest and Amortization

Excel in Business

Future Value

EXAMPLE

If for the next 40 years you invest Rs 20,000 each year towards your retirement and earn interest at the rate of 8% per annum on your investments, how much will you have upon retirement?

Future Value

Number of Years	**40**
Periodic Payment	Rs 20,000
Initial Balance	0
Annual Rate of Interest	8%
When do periodic payments occur?	Two cases: beginning or end of the year

Using FV Function

SYNTAX:

= FV (

rate per period, 8%

number of periods, 40

periodic payments, -20,000

initial balance, 0

[when do periodic payments occur] 0 or 1

)

0 for end of the period

1 for beginning of the period

Using FV function (cont…)

The FV function translates as

=FV (8%,40,-20000,0,1)

Remember: Payments made by you are always entered with a negative sign.

Present Value

Suppose you wish to buy a printer and there are two plans available:

1. Pay Rs 11,000 today

2. Pay Rs 3000 at the end of each year for the next 5 years

If the market rate of interest is 12%, which of the above options is a better deal?

Evaluating Plan 2

Number of Years	**5**
Periodic Payment	Rs 3000
Annual Rate of Interest	12%
When do periodic payments occur?	Two cases: beginning or end of the year

Using PV Function

SYNTAX:

= PV (

rate per period, 12%

number of periods, 5

periodic payments, -3000

cash balance to be maintained? 0 in our example

[when do additional payments occur] 0 (end)

)

Using PV function (cont…)

The PV function translates as

=PV (12%,5,-3000,0,0)

Remember: Payments made by you are always entered with a negative sign.

Building Amortization Tables

You want to pay back a loan of Rs 10,000 in 10 months.

What will be your monthly installment if the bank charges an interest of 8% per annum?

Building Amortization Tables

The installment P is given by

$$P = \frac{10{,}000}{\left(\frac{1}{1+r} + \frac{1}{(1+r)^2} + \ldots + \frac{1}{(1+r)^{10}} \right)}$$

which in Excel can be easily calculated using the PMT function.

PMT Function

SYNTAX:

= PMT (

rate per period, 8%/12 in our example

number of periods, 10 in our example

loan amount, 10,000 in our example

cash balance to be maintained? 0 in our example

[when do additional payments occur] 0 (end)

)

Using PMT function (cont…)

The PMT function translates as

=PMT (8%/12,10,10000,0,0)

Remember: Payments made to you are always entered with a positive sign.

The Amortization Table

	A	B	C	D	E	F
3	Loan Amount	10000				
4	Annual Rate	=8%				
5	Monthly Rate	=B4/12				
6						
7	Installment	=-PMT(B5,10,B3,0,0)				
8						
9	MONTH	Outstanding balance before the payment	Periodic Payment	Outstanding balance after the payment	Interest Repaid	Principal Repaid
10	0	10000	0	10000	0	0
11	1	=D10*(1+$B$5)	=$B$7	=B11-C11	=B11-D10	=C11-E11
12	2	=D11*(1+$B$5)	=$B$7	=B12-C12	=B12-D11	=C12-E12
13	3	=D12*(1+$B$5)	=$B$7	=B13-C13	=B13-D12	=C13-E13
14	4	=D13*(1+$B$5)	=$B$7	=B14-C14	=B14-D13	=C14-E14
15	5	=D14*(1+$B$5)	=$B$7	=B15-C15	=B15-D14	=C15-E15
16	6	=D15*(1+$B$5)	=$B$7	=B16-C16	=B16-D15	=C16-E16
17	7	=D16*(1+$B$5)	=$B$7	=B17-C17	=B17-D16	=C17-E17
18	8	=D17*(1+$B$5)	=$B$7	=B18-C18	=B18-D17	=C18-E18
19	9	=D18*(1+$B$5)	=$B$7	=B19-C19	=B19-D18	=C19-E19
20	10	=D19*(1+$B$5)	=$B$7	=B20-C20	=B20-D19	=C20-E20

Faster way to build Amortization Tables

You require two functions:

- **PPMT** – determines what part of the installment is principal
- **IPMT** – determines what part of the installment is interest

PPMT Function

SYNTAX:

= PPMT (

rate per period, 8%/12 in our example

which period?, various months 1-10 in our example

number of periods, 10 in our example

loan amount, 10,000 in our example

cash balance to be maintained? 0 in our example

[when do additional payments occur] 0 (end)

)

IPMT Function

SYNTAX:

= IPMT (

rate per period, 8%/12 in our example

which period?, various months 1-10 in our example

number of periods, 10 in our example

loan amount, 10,000 in our example

cash balance to be maintained? 0 in our example

[when do additional payments occur] 0 (end)

)

Using These Functions

	A	B	C	D	E	F
3	Loan Amount	10000				
4	Annual Rate	=8%/12				
5	Installment	=-PMT(B4,10,B3,0,0)				
6						
7	**MONTH**	**Outstanding balance before the payment**	**Periodic Payment**	**Outstanding balance after the payment**	**Principal Repaid**	**Interest Repaid**
8						
9	0	10000	0	=B9-C9	0	0
10	1	=FV(rate,1,-D9,0,1)	=$B$5	=B10-C10	=-PPMT(rate,A10,10,loan,0,0)	=-IPMT(rate,A10,10,loan,0,0)
11	2	=FV(rate,1,-D10,0,1)	=$B$5	=B11-C11	=-PPMT(rate,A11,10,loan,0,0)	=-IPMT(rate,A11,10,loan,0,0)
12	3	=FV(rate,1,-D11,0,1)	=$B$5	=B12-C12	=-PPMT(rate,A12,10,loan,0,0)	=-IPMT(rate,A12,10,loan,0,0)
13	4	=FV(rate,1,-D12,0,1)	=$B$5	=B13-C13	=-PPMT(rate,A13,10,loan,0,0)	=-IPMT(rate,A13,10,loan,0,0)
14	5	=FV(rate,1,-D13,0,1)	=$B$5	=B14-C14	=-PPMT(rate,A14,10,loan,0,0)	=-IPMT(rate,A14,10,loan,0,0)
15	6	=FV(rate,1,-D14,0,1)	=$B$5	=B15-C15	=-PPMT(rate,A15,10,loan,0,0)	=-IPMT(rate,A15,10,loan,0,0)
16	7	=FV(rate,1,-D15,0,1)	=$B$5	=B16-C16	=-PPMT(rate,A16,10,loan,0,0)	=-IPMT(rate,A16,10,loan,0,0)
17	8	=FV(rate,1,-D16,0,1)	=$B$5	=B17-C17	=-PPMT(rate,A17,10,loan,0,0)	=-IPMT(rate,A17,10,loan,0,0)
18	9	=FV(rate,1,-D17,0,1)	=$B$5	=B18-C18	=-PPMT(rate,A18,10,loan,0,0)	=-IPMT(rate,A18,10,loan,0,0)
19	10	=FV(rate,1,-D18,0,1)	=$B$5	=B19-C19	=-PPMT(rate,A19,10,loan,0,0)	=-IPMT(rate,A19,10,loan,0,0)
20						

Other Functions

RATE Function

EXAMPLE

I want to borrow Rs 80,000 and make monthly payments for 10 years. The maximum monthly payment that I can afford is Rs 1000.

What is the maximum rate of interest that I can bear?

RATE Function (Cont…)

SYNTAX:

= RATE (

number of periods, 120 in our example

periodic payments, -1000 in our example

loan amount 80000 in our example

cash balance 0 in our example

[when do payments occur] 0 (end)

)

RATE Function (Cont…)

The RATE function translates as

=RATE (120, -1000, 80000, 0, 0)

NPER Function

EXAMPLE

I borrow Rs 100,000 at 8% interest and make yearly payments of Rs 10,000.

How many years will it take to amortize the loan?

NPER Function (Cont…)

SYNTAX:

= NPER (

rate, 8% in our example

periodic payments, 10,000 in our example

loan amount 100,000 in our example

cash balance 0 in our example

[when do payments occur] 0 (end)

)

NPER Function (Cont…)

The RATE function translates as

=NPER (8%, -10000, 100000, 0, 0)